

- (a) Use a CAS to plot d'Alembert's solution (14) on $[-6, 6]$ for $t = 0.2k$, $k = 0, 1, 2, \dots, 25$. Assume that $a = 1$.
- (b) Use the animation feature of your computer algebra system to make a movie of the solution. Describe the motion of the string over time.
27. The model of the vibrating string in Problem 7 is called a **plucked string**.
- (a) Use a CAS to plot the partial sum $S_6(x, t)$; that is, the first six nonzero terms of your solution $u(x, t)$, for $t = 0.1k$, $k = 0, 1, 2, \dots, 20$. Assume that $a = 1$, $h = 1$, and $L = \pi$.
- (b) Use the animation feature of your computer algebra system to make a movie of the solution to Problem 7.
28. Consider the vibrating string in Problem 10. Use a CAS to plot the partial sum $S_6(x, t)$; that is, the first six nonzero terms of your solution $u(x, t)$ for $t = 0.25k$, $k = 0, 2, 3, 4, 6, 8, 10, 14$. Assume that $a = 1$, $h = 1$, and $L = \pi$. Then superimpose the eight graphs on the same coordinate system.

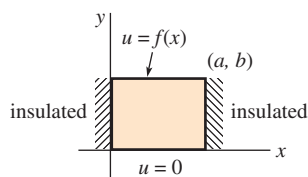


FIGURE 13.5.1 Find the temperature u in a rectangular plate

13.5 Laplace's Equation

INTRODUCTION Suppose we wish to find the steady-state temperature $u(x, y)$ in a rectangular plate whose vertical edges $x = 0$ and $x = a$ are insulated, and whose upper and lower edges $y = b$ and $y = 0$ are maintained at temperatures $f(x)$ and 0 , respectively. See **FIGURE 13.5.1**. When no heat escapes from the lateral faces of the plate, we solve the following boundary-value problem:

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0, \quad 0 < x < a, \quad 0 < y < b \quad (1)$$

$$\frac{\partial u}{\partial x} \Big|_{x=0} = 0, \quad \frac{\partial u}{\partial x} \Big|_{x=a} = 0, \quad 0 < y < b \quad (2)$$

$$u(x, 0) = 0, \quad u(x, b) = f(x), \quad 0 < x < a. \quad (3)$$

Solution of the BVP With $u(x, y) = X(x)Y(y)$, separation of variables in (1) leads to

$$\frac{X''}{X} = -\frac{Y''}{Y} = -\lambda$$

$$X'' + \lambda X = 0 \quad (4)$$

$$Y'' - \lambda Y = 0. \quad (5)$$

The three homogeneous boundary conditions in (2) and (3) translate into $X'(0) = 0$, $X'(a) = 0$, and $Y(0) = 0$. The Sturm–Liouville problem associated with the equation in (4) is then

$$X'' + \lambda X = 0, \quad X'(0) = 0, \quad X'(a) = 0. \quad (6)$$

Examination of the cases corresponding to $\lambda = 0$, $\lambda = -\alpha^2 < 0$, and $\lambda = \alpha^2 > 0$, where $\alpha > 0$, has already been carried out in Example 1 in Section 12.5. For convenience a shortened version of that analysis follows.

For $\lambda = 0$, (6) becomes

$$X'' = 0, \quad X'(0) = 0, \quad X'(a) = 0.$$

The solution of the ODE is $X = c_1 + c_2x$. The boundary condition $X'(0) = 0$ then implies $c_2 = 0$, and so $X = c_1$. Note that for any c_1 , this constant solution satisfies the second boundary condition $X'(a) = 0$. By imposing $c_1 \neq 0$, $X = c_1$ is a nontrivial solution of the BVP (6). For $\lambda = -\alpha^2 < 0$, (6) possesses no nontrivial solution. For $\lambda = \alpha^2 > 0$, (6) becomes

$$X'' + \alpha^2 X = 0, \quad X'(0) = 0, \quad X'(a) = 0.$$

Applying the boundary condition $X'(0) = 0$ the solution $X = c_1 \cos \alpha x + c_2 \sin \alpha x$ implies $c_2 = 0$ and so $X = c_1 \cos \alpha x$. The second boundary condition $X'(a) = 0$ applied to this last expression then gives $-c_1 \alpha \sin \alpha a = 0$. Because $\alpha > 0$, the last equation is satisfied when $\alpha a = n\pi$ or $\alpha = n\pi/a$, $n = 1, 2, \dots$. The eigenvalues of (6) are then λ_0 and $\lambda_n = \alpha_n^2 = n^2 \pi^2 / a^2$, $n = 1, 2, \dots$

By corresponding $\lambda_0 = 0$ with $n = 0$, the eigenfunctions of (6) are

$$X = c_1, \quad n = 0, \quad \text{and} \quad X = c_1 \cos \frac{n\pi}{a} x, \quad n = 1, 2, \dots$$

We must now solve equation (5) subject to the single homogeneous boundary condition $Y(0) = 0$. First, for $\lambda_0 = 0$ the DE in (5) is simply $Y'' = 0$, and thus its solution is $Y = c_3 + c_4 y$. But $Y(0) = 0$ implies $c_3 = 0$ so $Y = c_4 y$. Second, for $\lambda_n = n^2 \pi^2 / a^2$, the DE in (5) is $Y'' - \frac{n^2 \pi^2}{a^2} Y = 0$. Because $0 < y < b$ is a finite interval, we write the general solution in terms of hyperbolic functions:

$$Y(y) = c_3 \cosh(n\pi y/a) + c_4 \sinh(n\pi y/a).$$

From this solution we see $Y(0) = 0$ again implies $c_3 = 0$ so $Y = c_4 \sinh(n\pi y/a)$.

Thus product solutions $u_n = X(x)Y(y)$ that satisfy the Laplace's equation (1) and the three homogeneous boundary conditions in (2) and (3) are

$$A_0 y, \quad n = 0, \quad \text{and} \quad A_n \sinh \frac{n\pi}{a} y \cos \frac{n\pi}{a} x, \quad n = 1, 2, \dots,$$

where we have rewritten $c_1 c_4$ as A_0 for $n = 0$ and as A_n for $n = 1, 2, \dots$

The superposition principle yields another solution

$$u(x, y) = A_0 y + \sum_{n=1}^{\infty} A_n \sinh \frac{n\pi}{a} y \cos \frac{n\pi}{a} x. \quad (7)$$

Finally, by substituting $y = b$ in (7) we see

$$u(x, b) = f(x) = A_0 b + \sum_{n=1}^{\infty} \left(A_n \sinh \frac{n\pi}{a} b \right) \cos \frac{n\pi}{a} x,$$

is a half-range expansion of f in a Fourier cosine series. If we make the identifications $A_0 b = a_0/2$ and $A_n \sinh(n\pi b/a) = a_n$, $n = 1, 2, \dots$, it follows from (2) and (3) of Section 12.3 that

$$\begin{aligned} 2A_0 b &= \frac{2}{a} \int_0^a f(x) dx \\ A_0 &= \frac{1}{ab} \int_0^a f(x) dx \end{aligned} \quad (8)$$

and

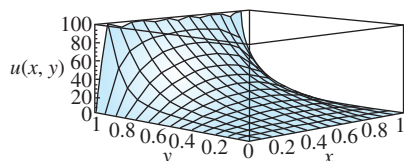
$$\begin{aligned} A_n \sinh \frac{n\pi}{a} b &= \frac{2}{a} \int_0^a f(x) \cos \frac{n\pi}{a} x dx \\ A_n &= \frac{2}{a \sinh \frac{n\pi}{a} b} \int_0^a f(x) \cos \frac{n\pi}{a} x dx. \end{aligned} \quad (9)$$

The solution of the boundary-value problem (1)–(3) consists of the series in (7), with coefficients A_0 and A_n defined in (8) and (9), respectively.

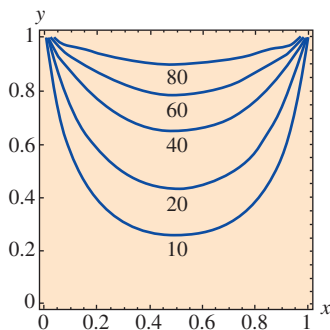
Dirichlet Problem A boundary-value problem in which we seek a solution to an elliptic partial differential equation such as Laplace's equation $\nabla^2 u = 0$ within a region R (in the plane or 3-space) such that u takes on prescribed values on the entire boundary of the region is called a **Dirichlet problem**. In Problem 1 in Exercises 13.5 you are asked to show that the solution of the Dirichlet problem for a rectangular region

$$\begin{aligned} \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} &= 0, \quad 0 < x < a, \quad 0 < y < b \\ u(0, y) &= 0, \quad u(a, y) = 0 \\ u(x, 0) &= 0, \quad u(x, b) = f(x) \end{aligned}$$

Why hyperbolic functions? 
See pages 122 and 692.



(a) Surface



(b) Isotherms

FIGURE 13.5.2 Surface is graph of partial sums when $f(x) = 100$ and $a = b = 1$ in (10)

is

$$u(x, y) = \sum_{n=1}^{\infty} A_n \sinh \frac{n\pi}{a} y \sin \frac{n\pi}{a} x \quad \text{where} \quad A_n = \frac{2}{a \sinh \frac{n\pi b}{a}} \int_0^a f(x) \sin \frac{n\pi}{a} x \, dx. \quad (10)$$

In the special case when $f(x) = 100$, $a = 1$, $b = 1$, the coefficients A_n are given by $A_n = 200 \frac{1 - (-1)^n}{n\pi \sinh n\pi}$. With the help of a CAS the plot of the surface defined by $u(x, y)$ over the region R : $0 \leq x \leq 1$, $0 \leq y \leq 1$ is given in **FIGURE 13.5.2(a)**. You can see in the figure that boundary conditions are satisfied; especially note that along $y = 1$, $u = 100$ for $0 \leq x \leq 1$. The isotherms, or curves, in the rectangular region along which the temperature $u(x, y)$ is constant can be obtained using the contour plotting capabilities of a CAS and are illustrated in Figure 13.5.2(b). The isotherms can also be visualized as the curves of intersection (projected into the xy -plane) of horizontal planes $u = 80$, $u = 60$, and so on, with the surface in Figure 13.5.2(a). Notice that throughout the region the maximum temperature is $u = 100$ and occurs on the portion of the boundary corresponding to $y = 1$. This is no coincidence. There is a **maximum principle** that states a solution u of Laplace's equation within a bounded region R with boundary B (such as a rectangle, circle, sphere, and so on) takes on its maximum and minimum values on B . In addition, it can be proved that u can have no relative extrema (maxima or minima) in the interior of R . This last statement is clearly borne out by the surface shown in Figure 13.5.2(a).

Superposition Principle A Dirichlet problem for a rectangle can be readily solved by separation of variables when homogeneous boundary conditions are specified on two *parallel* boundaries. However, the method of separation of variables is not applicable to a Dirichlet problem when the boundary conditions on all four sides of the rectangle are nonhomogeneous. To get around this difficulty we break the boundary-value problem

$$\begin{aligned} \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} &= 0, \quad 0 < x < a, \quad 0 < y < b \\ u(0, y) &= F(y), \quad u(a, y) = G(y), \quad 0 < y < b \\ u(x, 0) &= f(x), \quad u(x, b) = g(x), \quad 0 < x < a \end{aligned} \quad (11)$$

into two problems, each of which has homogeneous boundary conditions on parallel boundaries, as shown.

Problem 1	Problem 2
$\frac{\partial^2 u_1}{\partial x^2} + \frac{\partial^2 u_1}{\partial y^2} = 0, \quad 0 < x < a, \quad 0 < y < b$	$\frac{\partial^2 u_2}{\partial x^2} + \frac{\partial^2 u_2}{\partial y^2} = 0, \quad 0 < x < a, \quad 0 < y < b$
$u_1(0, y) = 0, \quad u_1(a, y) = 0, \quad 0 < y < b$	$u_2(0, y) = F(y), \quad u_2(a, y) = G(y), \quad 0 < y < b$
$u_1(x, 0) = f(x), \quad u_1(x, b) = g(x), \quad 0 < x < a$	$u_2(x, 0) = 0, \quad u_2(x, b) = 0, \quad 0 < x < a$

Suppose u_1 and u_2 are the solutions of Problems 1 and 2, respectively. If we define $u(x, y) = u_1(x, y) + u_2(x, y)$, it is seen that u satisfies all boundary conditions in the original problem (11). For example,

$$u(0, y) = u_1(0, y) + u_2(0, y) = 0 + F(y) = F(y)$$

$$u(x, b) = u_1(x, b) + u_2(x, b) = g(x) + 0 = g(x)$$

and so on. Furthermore, u is a solution of Laplace's equation by Theorem 13.1.1. In other words, by solving Problems 1 and 2 and adding their solutions we have solved the original

problem. This additive property of solutions is known as the superposition principle. See **FIGURE 13.5.3**.

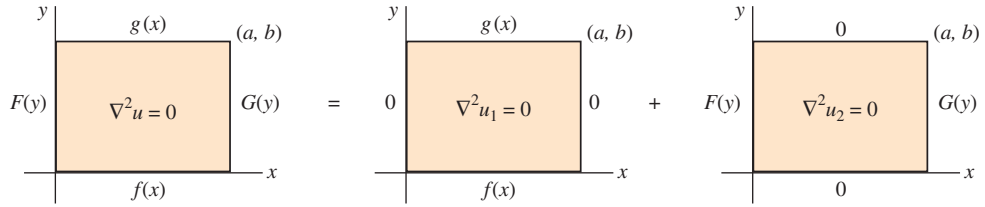


FIGURE 13.5.3 Solution u = Solution u_1 of Problem 1 + Solution u_2 of Problem 2

We leave as exercises (see Problems 13 and 14 in Exercises 13.5) to show that a solution of Problem 1 is

$$u_1(x, y) = \sum_{n=1}^{\infty} \left\{ A_n \cosh \frac{n\pi}{a} y + B_n \sinh \frac{n\pi}{a} y \right\} \sin \frac{n\pi}{a} x,$$

where
$$A_n = \frac{2}{a} \int_0^a f(x) \sin \frac{n\pi}{a} x \, dx$$

$$B_n = \frac{1}{\sinh \frac{n\pi}{a} b} \left(\frac{2}{a} \int_0^a g(x) \sin \frac{n\pi}{a} x \, dx - A_n \cosh \frac{n\pi}{a} b \right),$$

and that a solution of Problem 2 is

$$u_2(x, y) = \sum_{n=1}^{\infty} \left\{ A_n \cosh \frac{n\pi}{b} x + B_n \sinh \frac{n\pi}{b} x \right\} \sin \frac{n\pi}{b} y,$$

where
$$A_n = \frac{2}{b} \int_0^b F(y) \sin \frac{n\pi}{b} y \, dy$$

$$B_n = \frac{1}{\sinh \frac{n\pi}{b} a} \left(\frac{2}{b} \int_0^b G(y) \sin \frac{n\pi}{b} y \, dy - A_n \cosh \frac{n\pi}{b} a \right).$$

13.5

Exercises

Answers to selected odd-numbered problems begin on page ANS-32.

In Problems 1–10, solve Laplace's equation (1) for a rectangular plate subject to the given boundary conditions.

1. $u(0, y) = 0$, $u(a, y) = 0$

$u(x, 0) = 0$, $u(x, b) = f(x)$

2. $u(0, y) = 0$, $u(a, y) = 0$

$\frac{\partial u}{\partial y} \Big|_{y=0} = 0$, $u(x, b) = f(x)$

3. $u(0, y) = 0$, $u(a, y) = 0$

$u(x, 0) = f(x)$, $u(x, b) = 0$

4. $\frac{\partial u}{\partial x} \Big|_{x=0} = 0$, $\frac{\partial u}{\partial x} \Big|_{x=a} = 0$

$u(x, 0) = x$, $u(x, b) = 0$

5. $u(0, y) = 0$, $u(1, y) = 1 - y$

$\frac{\partial u}{\partial y} \Big|_{y=0} = 0$, $\frac{\partial u}{\partial y} \Big|_{y=1} = 0$

6. $u(0, y) = g(y)$, $\frac{\partial u}{\partial x} \Big|_{x=1} = 0$

$\frac{\partial u}{\partial y} \Big|_{y=0} = 0$, $\frac{\partial u}{\partial y} \Big|_{y=\pi} = 0$

7. $\frac{\partial u}{\partial x} \Big|_{x=0} = u(0, y)$, $u(\pi, y) = 1$

$u(x, 0) = 0$, $u(x, \pi) = 0$

8. $u(0, y) = 0$, $u(1, y) = 0$

$\frac{\partial u}{\partial y} \Big|_{y=0} = u(x, 0)$, $u(x, 1) = f(x)$

9. $u(0, y) = 0$, $u(1, y) = 0$
 $u(x, 0) = 100$, $u(x, 1) = 200$
10. $u(0, y) = 10y$, $\left. \frac{\partial u}{\partial x} \right|_{x=1} = -1$
 $u(x, 0) = 0$, $u(x, 1) = 0$

In Problems 11 and 12, solve Laplace's equation (1) for the semi-infinite plate extending in the positive y -direction. In each case assume that $u(x, y)$ is bounded at $y \rightarrow \infty$.

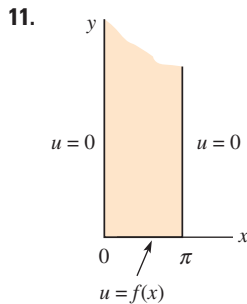


FIGURE 13.5.4 Semi-infinite Plate in Problem 11

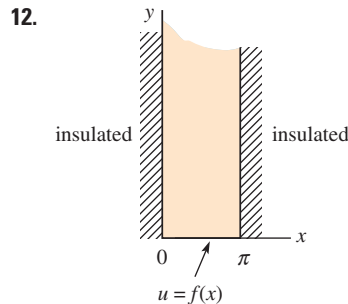


FIGURE 13.5.5 Semi-infinite Plate in Problem 12

In Problems 13 and 14, solve Laplace's equation (1) for a rectangular plate subject to the given boundary conditions.

13. $u(0, y) = 0$, $u(a, y) = 0$
 $u(x, 0) = f(x)$, $u(x, b) = g(x)$
14. $u(0, y) = F(y)$, $u(a, y) = G(y)$
 $u(x, 0) = 0$, $u(x, b) = 0$

In Problems 15 and 16, use the superposition principle to solve Laplace's equation (1) for a square plate subject to the given boundary conditions.

15. $u(0, y) = 1$, $u(\pi, y) = 1$
 $u(x, 0) = 0$, $u(x, \pi) = 1$
16. $u(0, y) = 0$, $u(2, y) = y(2 - y)$
 $u(x, 0) = 0$, $u(x, 2) = \begin{cases} x, & 0 < x < 1 \\ 2 - x, & 1 \leq x < 2 \end{cases}$
17. In Problem 16, what is the maximum value of the temperature u for $0 \leq x \leq 2$, $0 \leq y \leq 2$?

Computer Lab Assignments

18. (a) In Problem 1 suppose $a = b = \pi$ and $f(x) = 100x(\pi - x)$. Without using the solution $u(x, y)$ sketch, by hand, what the surface would look like over the rectangular region defined by $0 \leq x \leq \pi$, $0 \leq y \leq \pi$.
- (b) What is the maximum value of the temperature u for $0 \leq x \leq \pi$, $0 \leq y \leq \pi$?
- (c) Use the information in part (a) to compute the coefficients for your answer in Problem 1. Then use the 3D-plot application of your CAS to graph the partial sum $S_5(x, y)$ consisting of the first five nonzero terms of the solution in part (a) for $0 \leq x \leq \pi$, $0 \leq y \leq \pi$. Use different perspectives and then compare with part (a).

19. (a) Use the contour-plot application of your CAS to graph the isotherms $u = 170, 140, 110, 80, 60, 30$ for the solution of Problem 9. Use the partial sum $S_5(x, y)$ consisting of the first five nonzero terms of the solution.
- (b) Use the 3D-plot application of your CAS to graph the partial sum $S_5(x, y)$.
20. Use the contour-plot application of your CAS to graph the isotherms $u = 2, 1, 0.5, 0.2, 0.1, 0.05, 0, -0.05$ for the solution of Problem 10. Use the partial sum $S_5(x, y)$ consisting of the first five nonzero terms of the solution.

Discussion Problems

21. Solve the Neumann problem for a rectangle:

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0, \quad 0 < x < a, \quad 0 < y < b$$

$$\left. \frac{\partial u}{\partial y} \right|_{y=0} = 0, \quad \left. \frac{\partial u}{\partial y} \right|_{y=b} = 0, \quad 0 < x < a$$

$$\left. \frac{\partial u}{\partial x} \right|_{x=0} = 0, \quad \left. \frac{\partial u}{\partial x} \right|_{x=a} = g(y), \quad 0 < y < b.$$

- (a) Explain why a necessary condition for a solution u to exist is that g satisfy

$$\int_0^b g(y) dy = 0.$$

This is sometimes called a **compatibility condition**. Do some extra reading and explain the compatibility condition on physical grounds.

- (b) If u is a solution of the BVP, explain why $u + c$, where c is an arbitrary constant, is also a solution.

22. Consider the boundary-value problem

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0, \quad 0 < x < 1, \quad 0 < y < \pi$$

$$u(0, y) = u_0 \cos y, \quad u(1, y) = u_0(1 + \cos 2y)$$

$$\left. \frac{\partial u}{\partial y} \right|_{y=0} = 0, \quad \left. \frac{\partial u}{\partial y} \right|_{y=\pi} = 0.$$

Discuss how the following answer was obtained

$$u(x, y) = u_0 x + \frac{u_0}{\sinh 1} \sinh(1 - x) \cos y + \frac{u_0}{\sinh 2} \sinh 2x \cos 2y.$$

Carry out your ideas.